TSM 098 | INDEPENDENT PRACTICE

Adding and subtracting numerical fractions

While TSM 099 studies Topic 23. Review: Topic 21, Topic 22.

Name: _____ Date: _____

Complete the core pages first; use the extension page for remaining time. Show all work here or on separate paper. Check answers after attempting each section and correct errors.

Core: adding fractions

1. Evaluate and simplify.

$$\frac{5}{12} + (\frac{2}{2})$$

2. Evaluate and simplify.

$$\frac{-5}{6} + (\frac{7}{6})$$

3. Evaluate and simplify.

$$\frac{6}{4} + (\frac{-4}{4})$$

4. Evaluate and simplify.

$$\frac{8}{3} + (\frac{4}{6})$$

5. Evaluate and simplify.

$$\frac{-8}{3} + (\frac{2}{2})$$

6. Evaluate and simplify.

$$\frac{8}{6} + (\frac{1}{6})$$

7. Evaluate and simplify.

$$\frac{7}{5} + (\frac{-1}{3})$$

8. Evaluate and simplify.

$$\frac{-1}{8} + (\frac{6}{8})$$

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Core: subtracting fractions

9. Evaluate and simplify.

$$\frac{3}{8} - (\frac{4}{8})$$

10. Evaluate and simplify.

$$\frac{-6}{8} - (\frac{6}{6})$$

11. Evaluate and simplify.

$$\frac{5}{10} - (\frac{-6}{6})$$

12. Evaluate and simplify.

$$\frac{2}{12} - (\frac{8}{3})$$

13. Evaluate and simplify.

$$\frac{-8}{8} - (\frac{8}{8})$$

14. Evaluate and simplify.

$$\frac{3}{4} - (\frac{3}{6})$$

15. Evaluate and simplify.

$$\frac{6}{10} - (\frac{-9}{3})$$

16. Evaluate and simplify.

$$\frac{-2}{3} - (\frac{2}{5})$$

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Complete the core pages first; use the extension page for remaining time. Show all work here or on separate paper. Check answers after attempting each section and correct errors.

Core: mixed addition and subtraction

17. Evaluate and simplify.

$$\frac{1}{8} + (\frac{5}{6})$$

18. Evaluate and simplify.

$$\frac{-5}{12} + (\frac{6}{2})$$

19. Evaluate and simplify.

$$\frac{6}{5} + (\frac{-9}{4})$$

20. Evaluate and simplify.

$$\frac{8}{8} + (\frac{4}{4})$$

21. Evaluate and simplify.

$$\frac{3}{3} - (\frac{2}{8})$$

22. Evaluate and simplify.

$$\frac{-8}{4} - (\frac{6}{6})$$

23. Evaluate and simplify.

$$\frac{3}{10} - (\frac{-2}{5})$$

24. Evaluate and simplify.

$$\frac{3}{6} - (\frac{1}{2})$$

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Name: _____ Date: _____

Complete the core pages first; use the extension page for remaining time. Show all work here or on separate paper. Check answers after attempting each section and correct errors.

Extension: longer expressions and reasoning

25. Evaluate and simplify. Show intermediate steps.

$$1+(-\frac{4}{5})-(-\frac{3}{4})$$

26. Evaluate and simplify. Show intermediate steps.

$$-\frac{5}{3}+(-\frac{1}{2})-(\frac{1}{4})$$

27. Evaluate and simplify. Show intermediate steps.

$$-\frac{5}{4}+(-\frac{2}{3})-(\frac{1}{2})$$

28. Evaluate and simplify. Show intermediate steps.

$$-\frac{5}{3}+(\frac{1}{2})-(1)$$

29. Evaluate and simplify. Show intermediate steps.

$$1+(\frac{2}{5})-(-\frac{3}{4})$$

30. Evaluate and simplify. Show intermediate steps.

$$-\frac{1}{2}+(\frac{1}{3})-(-1)$$

31. Explain the error and give the correct answer.

$$\frac{1}{3}+\frac{1}{4}=\frac{2}{7}$$

32. Evaluate. Explain why the sign of the answer makes sense.

$$\frac{2}{5}-\frac{7}{10}$$

TSM 098 | ANSWER KEY

Adding and subtracting numerical fractions

While TSM 099 studies Topic 23. Review: Topic 21, Topic 22.

Equivalent forms are acceptable. Use these answers after attempting the problems; revisit the linked lecture notes for additional examples.

Core: adding fractions

1. Evaluate and simplify.

$$\frac{17}{12} = \frac{17}{12}$$

2. Evaluate and simplify.

$$\frac{2}{6} = \frac{1}{3}$$

3. Evaluate and simplify.

$$\frac{2}{4} = \frac{1}{2}$$

4. Evaluate and simplify.

$$\frac{20}{6} = \frac{10}{3}$$

5. Evaluate and simplify.

$$\frac{-10}{6} = -\frac{5}{3}$$

6. Evaluate and simplify.

$$\frac{9}{6} = \frac{3}{2}$$

7. Evaluate and simplify.

$$\frac{16}{15} = \frac{16}{15}$$

8. Evaluate and simplify.

$$\frac{5}{8} = \frac{5}{8}$$

Adding and subtracting numerical fractions

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Equivalent forms are acceptable. Use these answers after attempting the problems; revisit the linked lecture notes for additional examples.

Core: subtracting fractions

9. Evaluate and simplify.

$$\frac{-1}{8} = -\frac{1}{8}$$

10. Evaluate and simplify.

$$\frac{-42}{24} = -\frac{7}{4}$$

11. Evaluate and simplify.

$$\frac{45}{30} = \frac{3}{2}$$

12. Evaluate and simplify.

$$\frac{-30}{12} = -\frac{5}{2}$$

13. Evaluate and simplify.

$$\frac{-16}{8} = -2$$

14. Evaluate and simplify.

$$\frac{3}{12} = \frac{1}{4}$$

15. Evaluate and simplify.

$$\frac{108}{30} = \frac{18}{5}$$

16. Evaluate and simplify.

$$\frac{-16}{15} = -\frac{16}{15}$$

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Adding and subtracting numerical fractions

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Equivalent forms are acceptable. Use these answers after attempting the problems; revisit the linked lecture notes for additional examples.

Core: mixed addition and subtraction

17. Evaluate and simplify.

$$\frac{23}{24} = \frac{23}{24}$$

18. Evaluate and simplify.

$$\frac{31}{12} = \frac{31}{12}$$

19. Evaluate and simplify.

$$\frac{-21}{20} = -\frac{21}{20}$$

20. Evaluate and simplify.

$$\frac{16}{8} = 2$$

21. Evaluate and simplify.

$$\frac{18}{24} = \frac{3}{4}$$

22. Evaluate and simplify.

$$\frac{-36}{12} = -3$$

23. Evaluate and simplify.

$$\frac{7}{10} = \frac{7}{10}$$

24. Evaluate and simplify.

$$\frac{0}{6} = 0$$

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Equivalent forms are acceptable. Use these answers after attempting the problems; revisit the linked lecture notes for additional examples.

Extension: longer expressions and reasoning

25. Evaluate and simplify. Show intermediate steps.

$$\frac{19}{20}$$

26. Evaluate and simplify. Show intermediate steps.

$$-\frac{29}{12}$$

27. Evaluate and simplify. Show intermediate steps.

$$-\frac{29}{12}$$

28. Evaluate and simplify. Show intermediate steps.

$$-\frac{13}{6}$$

29. Evaluate and simplify. Show intermediate steps.

$$\frac{43}{20}$$

30. Evaluate and simplify. Show intermediate steps.

$$\frac{5}{6}$$

31. Explain the error and give the correct answer.

$$\frac{4}{12}+\frac{3}{12}=\frac{7}{12}$$

Use a common denominator; do not add the denominators.

32. Evaluate. Explain why the sign of the answer makes sense.

$$\frac{4}{10}-\frac{7}{10}=-\frac{3}{10}$$

The amount subtracted is larger than the starting amount.